# NT-Net: A Semantic Segmentation Network for Extracting Lake Water Bodies From Optical Remote Sensing Images Based on Transformer

Hai-Feng Zhong, Qing Sun, Hong-Mei Sun, and Rui-Sheng Jia

***Abstract*— The automatic extraction of lake water is one of the research hotspots in the field of remote sensing image processing. Due to the small interclass variance between lakes and other ground objects, and the complex texture characteristics of lake boundaries, existing methods often have problems such as over-segmentation and inaccurate boundary segmentation when segmenting lake water bodies. To alleviate these problems, this article designs an end-to-end semantic segmentation network [noise-canceling transformer network (NT-Net)] for the automatic extraction of lake water bodies from remote sensing images. Aiming at the problem of over-segmentation caused by nonlake objects, an interference attenuation module is designed in the network. This module can model the key features that are distinguishable and suitable for segmenting lake water by analyzing the difference in feature representation between lakes and other ground objects, thus suppressing the feature representation of nonlake objects. To more accurately segment the lake boundary, a multilevel transformer module is designed. This module can capture the context association of boundary information and enhance the feature representation of boundary information by using the self-attention mechanism. The comparative experimental results show that, compared with the current mainstream semantic segmentation networks, the method in this article has advantages in extracting lake water bodies comprehensively and coherently.**

***Index Terms*— Convolutional neural network (CNN), lake segmentation, optical remote sensing images, semantic segmentation, transformer.**

## I. Introduction

THE rapid development of remote sensing technology has provided finer and higher resolution optical remote sensing images [1]. Segmenting lake water bodies from optical remote sensing images can be widely used in water area change detection, water resource protection, global climate change research, and flood disaster emergency response [2], [3], [4], [5]. Therefore, how accurately extracting lake water bodies in remote sensing images is the primary task in exploring its application value. The current automatic extraction of lake water faces the following challenges.



Fig. 1. Remote sensing image of lake water bodies. The red boxes represent disturbed areas similar to lake water bodies. Green boxes represent tiny tributaries at the boundary of the lake water bodies.

1) There are many surface objects similar to lake water bodies in remote sensing images, such as grass, jungle, and silt beside the lake. As shown by the red box in Fig. 1, the lake water body at the boundary exhibits similar spectral characteristics to the surrounding soil due to its shallow depth. This similarity between nonuniform objects results in a small interclass variance between lake water bodies and other ground objects.
2) In addition to the over-segmentation problem, how to accurately segment the boundaries of lake water bodies is also a difficult problem to solve. For example, as shown in the green box in Fig. 1, under the impact of lake water on the soil on the shore, a large number of capillary lake water bodies will appear at the boundary. Since these lake water bodies are too small in scale and do not have obvious texture features, they are difficult to identify.

In response to the above problems, this article designs a noise-canceling transformer network (NT-Net) to use the image's local and global information to achieve accurate extraction of lakes. NT-Net adopts an asymmetric "encode–decode" structure as a shared backend. Specifically, to address the over-segmentation problem caused by nonlake water objects, an interference attenuation module (IAM) is designed. The IAM can not only use the positional relationship of features to model the differential features between lake water and other objects, but also adaptively give more feature weights

Manuscript received 6 June 2022; revised 12 July 2022; accepted 27 July 2022. Date of publication 8 August 2022; date of current version 19 August 2022. This work was supported by the Humanities and Social Science Fund of the Ministry of Education of the People's Republic of China under Grant 21YJAZH077. *(Corresponding authors: Hong-Mei Sun; Rui-Sheng Jia.)*

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Digital Object Identifier 10.1109/TGRS.2022.3197402

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to the lake water by using the dependencies between feature channels. To improve the segmentation accuracy (Acc) of the lake boundary degree, the multilevel transformer (MT) module is designed. On the one hand, the MT can integrate multiscale contextual information about boundary features to preserve the integrity and continuity of boundary contours by modeling long-range semantic dependencies of boundary information. On the other hand, this module can use high-level semantic features to assist the modeling process of low-level texture features, thus mapping out more accurate boundary details. Unlike existing transformers, the MT can model the feature output of each downsampling stage, thus learning multiscale features that preserve spatial localization and optimize discontinuous lake boundaries. The main contributions of this article are as follows.

1) We designed a new semantic segmentation network NT-Net based on convolutional neural network (CNN) and transformer. NT-Net makes full use of CNN's ability to capture local detail information and transformer's ability to model long-range information, effectively improving the segmentation Acc of lake water bodies.
2) An IAM is designed in NT-Net, which effectively alleviates the problem of over-segmentation caused by nonlake objects by deeply mining the feature differences between lake water bodies and other ground objects.
3) An MT module that can focus on building global dependencies is designed. This module can perform multilevel global self-attention modeling on feature information to accurately express the continuous positional relationship between boundary information.

The remainder of this article is organized as follows. Section II discusses the previous work on lake segmentation. Section III introduces the detailed structure of the NT-Net. In Section IV, we illustrate the implementation steps of the experiment and analyze the experimental results in detail. Section V discusses the rationality of some parameters in NT-Net. Finally, Section VI summarizes our conclusions.

## II. Related Work

Traditional lake segmentation methods generally use threshold segmentation [6], clustering [7], and support vector machines [8] to extract lake water bodies in remote sensing images. For example, McFeeters [9] proposed a lake water segmentation method based on the normalized difference water index (NDWI) by using the band reflection of surface objects and achieved relatively ideal segmentation results. Obida *et al.* [10] proposed a segmentation method of river water using K-means clustering technique. This method divides the river area and the background area based on the shallow features of the lake water bodies. However, the above methods cannot solve the problem of vegetation and soil noise because it is difficult to choose an appropriate threshold when the lake and background are mixed in pixels. To solve noise interference in the segmentation process, Wu *et al.* [11] proposed a lake segmentation method based on support vector machines. This method builds a mathematical model for noise reduction by analyzing the characteristics of noise, thus reducing the negative impact of noise information. Although the Acc of traditional methods in extracting lake water bodies continues to improve, most of them use low-level features such as the color, shape, and texture of lake water bodies to complete the segmentation process. Since the deep features of lake water cannot be used to refine the segmentation results, the segmentation Acc is low.

CNNs have been shown to be effective in capturing both shallow detailed features and deep abstract features in images [12]. To make full use of the deep abstract features, researchers try to apply CNN to the segmentation task of lake water bodies to accurately segment lakes. For example, Chen [13] used CNN to segment water bodies in cases such as narrow lakes and wide lakes, and demonstrated that deep learning methods were more accurate than traditional methods when segmenting lakes in images. However, this method cannot perceive global contextual information, so the segmentation results have a low degree of continuity. To improve the perception of global information to achieve dense prediction of lake pixels, Hu *et al.* [14] designed a multibranch aggregation module using dilated convolutions. The module uses dilated convolutions with different dilation rates to sample the given input, so that the network can establish the correspondence between local pixels and global pixels, thus protecting the integrity of lake water. Network models that use similar structures include MSLWE-Net [15] and PA-U-Net [16]. Although the above methods can better perceive the context distribution of image information, they cannot accurately locate the boundary information of lake water. To segment the lake boundary more accurately, some methods try to use the attention mechanism [17], [18] in the network model to give more attention to the boundary information adaptively. For example, MEC-Net [19], DAU-Net [20], FWE-Net [21], and MIE-Net [22] all use compression and excitation operations to redistribute feature weights for ground information, thus focusing more on feature channels containing boundary information. In addition, some methods [23], [24] also try to use loss functions that can refine the boundary modeling process. For example, Miao *et al.* [25] used the edge-weighted loss to assign more weights to the pixels near the boundary of the lake water body, so that the network could pay more attention to the boundary information during the decoding process, thus improving the segmentation Acc of the boundary.

The above methods only capture the detailed features of the boundary from the local scope and do not use the contextual information of the boundary to improve the coherence of the contour. Inspired by the great success of Visio transformer (ViT) [26] in the field of image classification, some researchers have tried to exploit the self-attention mechanism [27], [28], [29] to obtain contextual associations of boundary features. For example, BA-Net [30] utilizes a transformer similar to ViT to capture informative long-range relationships and obtain more coherent boundary results. However, transformer involves complex matrix operations, so the computational complexity of BA-Net is high. In order to improve the operating efficiency of the algorithm, Gao *et al.* [31] designed STransFuse using Swin transformer [32]. STransFuse shifts

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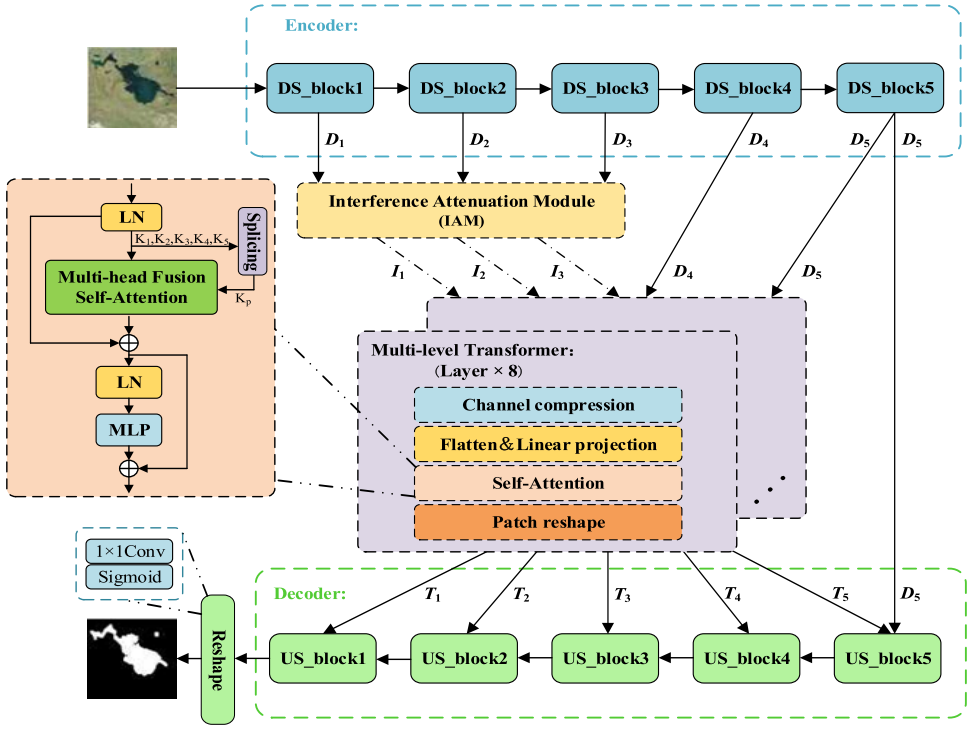



Fig. 2. Overview of NT-Net. The workflow of NT-Net is divided into two parts: encoding and self-attention process, and decoding process.

the focus of self-attention from global information to local information within each patch, thus reducing the computational complexity of the model. Xu *et al.* [33] proposed a network model for remote sensing image segmentation by transforming the backbone structure of the Swin transformer. The network uses depthwise separable convolutions [34] and a more flexible image window, which further reduces the computational complexity of the model. Although the above methods can reproduce the boundary contours of lakes well, these methods cannot identify the detailed texture of lake boundaries due to the weak inductive bias ability of the transformer. Inspired by the above work, we use CNN and transformer as the network's backbone to design a network model that can accurately extract lake water.

## III. NT-Net

### A. NT-Net Semantic Segmentation Network

To solve the problem of over-segmentation of lake water and accurately segment the boundary of lakes, this article designs a deep neural network NT-Net based on semantic segmentation. The network is mainly composed of encoder, decoder, IAM, and MT module, and its structure is shown in Fig. 2.

*1) Encoding and Self-Attention Process:* The encoder processes the input remote sensing images $I \in \mathbb{R}^{H\times W\times C}$ with successive downsampling blocks (DS-blocks) to generate a multiscale feature stream $D_i (i = 1, 2, 3, 4, 5)$. Before $D_i$ is input into the MT for self-attention processing, $D_1$, $D_2$, and $D_3$ need to be sent to IAM to reduce the negative impact of nonlake water objects on the segmentation process and generate feature streams $I_1$, $I_2$, and $I_3$. Considering that most of the interference generated by nonlake water objects exists in the low-level feature maps, only $D_1$, $D_2$, and $D_3$ are processed by IAM here. After that, the MT performs feature compression, flattening, and serialization on the input $I_1$, $I_2$, $I_3$, $D_4$, and $D_5$ to generate feature sequences for the self-attention process. Then, the multihead fusion attention mechanism (MF-SA) processes these feature sequences, thus completing the global self-attention process for features at different levels.

The reshaped feature stream $T_i (i = 1, 2, 3, 4, 5)$ can be sent to the decoder. The decoder uses the upsampling block (US_block) to upsample $T_i$ to restore the detailed features of the lake water gradually. Finally, the binary segmentation image of the lake water bodies is output after sigmoid processing.

*2) Decoding Process:* First, it is necessary to reshape the feature sequence generated by the self-attention process. This is because the output of the self-attention process is a 1-D feature sequence, which cannot be directly upsampled. The patch reshape operation is required to restore the dimension of the feature sequence to a 2-D size.

### B. Encoder

In NT-Net, the function of the encoder is to use the DS-block to extract the feature information of multiple scales from the remote sensing image for the self-attention process of the MT. The structure of the DS-block is shown in Fig. 3.

DS-block consists of a convolutional layer and a pooling layer, whose input feature stream is $\mathrm{FE}_i \in \mathbb{R}^{H\times W\times C_n}$ and the output feature stream is $\mathrm{FE}_{i+1} \in \mathbb{R}^{(H/2)\times(W/2)\times C_m}$, where $n < m$. The convolution layer draws on the idea of residual structure and improves the information diversity in the process

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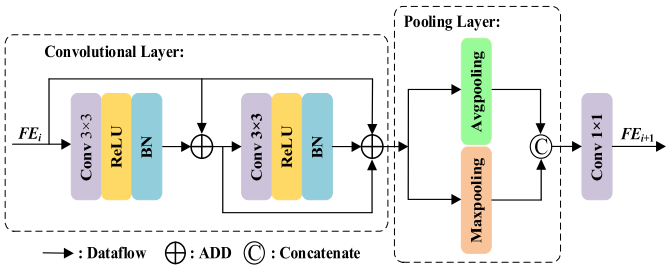



Fig. 3. Structure of the proposed DS-block.

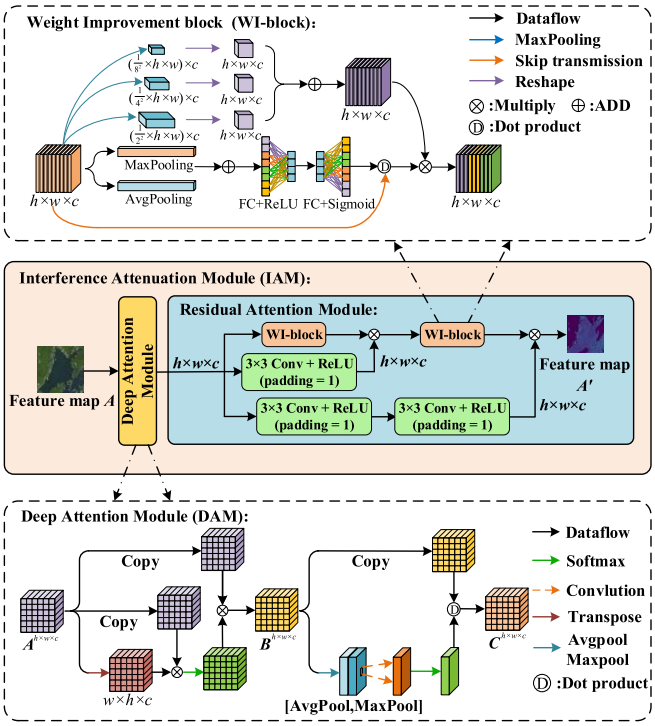



Fig. 4. Structure of the proposed IAM, which consists of DAM and RAM.

of feature fusion by fusing feature elements of different levels so that the features of the previous layer are fully utilized. The function of the pooling layer is to reduce the dimension of the extracted feature information so that the subsequent DS-block can obtain a larger receptive field, thus extracting deep-level semantic features. To enable CNN to extract rich texture information and contour information, we use average pooling and max pooling in the pooling layer.

### C. Interference Attenuation Module

To alleviate the over-segmentation problem caused by nonlake water objects, an IAM is designed, the structure of which is shown in Fig. 4. The innovation of IAM is that it can not only model the semantic representation of key features from spatial and channel dimensions but also use a variety of representative global features to enhance key features. IAM consists of a deep attention module (DAM) and a residual attention module (RAM). Among them, the role of the DAM is to use the positional relationship to model the differential features between lakes and other ground objects. The role of the RAM is to give more weight to the key features along the feature channel direction and utilize multiscale global information to emphasize key information, thus diluting the feature representation of the interference area. The workflow of the DAM and the RAM will be described in detail in the following.

*1) Deep Attention Module:* First, the initial feature maps $A \in \mathbb{R}^{h\times w\times c}$ need to be transposed to generate transposed feature maps $A^{\mathrm{T}} \in \mathbb{R}^{w\times h\times c}$. After the transposition operation is completed, the $A^{\mathrm{T}}$ needs to be multiplied with the $A$, and the product result is sent to the Softmax layer. It is worth noting that $A$ and $A^{\mathrm{T}}$ can be multiplied only when the $H$ and $W$ of the image are the same. If the $H$ and $W$ of the image are not the same, consider introducing data preprocessing operations to keep $H$ and $W$ consistent. Then, the output of the Softmax layer is multiplied back to the $A$ to generate the intermediate layer feature maps $B \in \mathbb{R}^{h\times w\times c}$. The purpose of the above steps is to model the feature differences between lake water bodies and other ground objects based on the spatial positional relationship of information. After completing the modeling of the differential features, it is necessary to use the pooling operation and the convolution operation to extract the representative lake water features from the $B$. Then, use the Softmax layer to reassign the weight ratio of the differential features and use the dot product operation to assign the reassigned information weight to the $B$, thus generating the feature maps $C \in \mathbb{R}^{h\times w\times c}$ that contain less disturbing information.

*2) Residual Attention Module:* To further suppress the feature representation of nonlake objects, a RAM is set up. This module mainly utilizes two consecutive weight improvement blocks (WI-blocks) to process the input feature maps $C \in \mathbb{R}^{h\times w\times c}$. WI-block can learn the nonlinear interaction between channels through squeezing and excitation operations, thus strengthening the differential performance of key information and interference information on the channel. In addition to using global pooling and average pooling, WI-block also uses global information at various scales such as $(h/2) \times (w/2) \times c$, $(h/4) \times (w/4) \times c$, and $(h/8) \times (w/8) \times c$ to improve the sensitivity to key information on the feature channels. To enhance the learning ability of the WI-block for discriminative features, we use the skip residual [35], [36] containing $3 \times 3$ convolutions to fuse the feature maps $C$ with the output of WI-block.

### D. Multilevel Transformer

The role of the MT module is to capture the global dependencies used to extract lake water information and improve the coherence of lake water boundaries. This module is capable of self-attention modeling of multiscale features of different paths to obtain rich boundary information, and its structure is shown in Fig. 5.

Compared with the MT, most of the existing methods [37], [38] insert the transformer as a single module into the encoder of the semantic segmentation network. Although this combination method is simple to operate, it will limit the performance of the transformer. Because the composition

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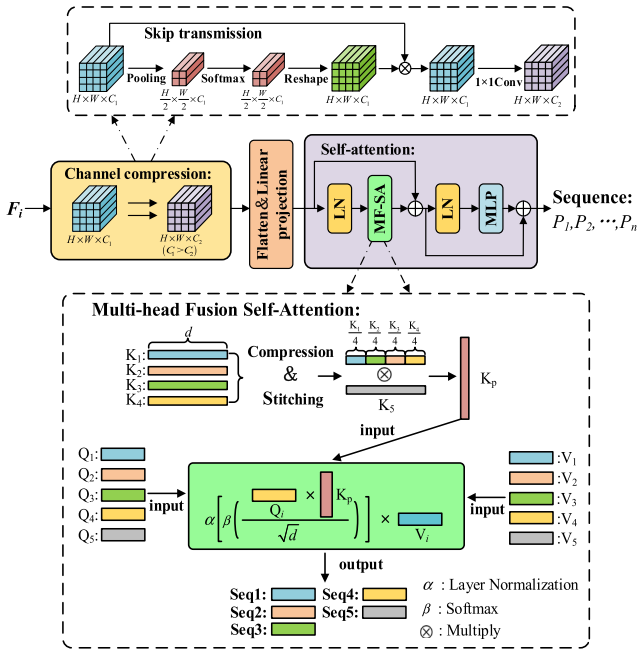



Fig. 5. Structure of the proposed MT module.

of images contains multiple levels of semantic features, this combination method can only model the global context of features at one level. Our research found that the initial images will generate feature maps with different semantic levels after passing through the CNN. If self-attention modeling is performed on feature maps at different levels, the spatial features and contextual semantic associations of target objects can be better captured.

Compared with the MT, most of the existing methods [37], [38] insert the transformer as a single module into the encoder of the semantic segmentation network. Although this combination method is simple to operate, it will limit the performance of the transformer. Because the composition of images contains multiple levels of semantic features, this combination method can only model the global context of features at one level. Our research found that the initial images will generate feature maps with different semantic levels after passing through the CNN. If self-attention modeling is performed on feature maps at different levels, the spatial features and contextual semantic associations of target objects can be better captured.

The novelty of the transformer is that it can perceive the distribution of lake boundaries in the global domain by modeling multiple levels of feature information without adding too many model parameters. In addition, it can use high-level semantic features to guide the modeling process of low-level texture features, thus improving the network's ability to identify detailed local features. The execution steps of the MT include channel compression and self-attention.

*1) Channel Compression:* The role of channel compression is to reduce useless feature channels to reduce the number of parameters used for transformer calculations. The input $F_i \in \mathbb{R}^{H_i \times W_i \times C_i} (i = 1, 2, 3, 4, 5)$ of this process is the $I_1$, $I_2$, and $I_3$ from the IAM and the $D_4$ and $D_5$ from the encoder, respectively. The number of channels of $I_1$, $I_2$, $I_3$, $D_4$, and $D_5$ is 64, 128, 256, 512, and 1024 in sequence.

First, the $F_i$ needs to undergo a pooling operation to extract the texture and contour information of the lake water bodies. The Softmax function and linear interpolation will reshape this part of the information to generate the key feature maps $F_i' \in \mathbb{R}^{H_i \times W_i \times C_i} (i = 1, 2, 3, 4, 5)$. Then, $F'$ is multiplied with the $F_i$ to generate the feature maps $E_i \in \mathbb{R}^{H_i \times W_i \times C_i} (i = 1, 2, 3, 4, 5)$ that contain rich lake's information. Finally, a $1 \times 1$ convolution kernel is utilized to extract the useful key information in $E_i$ and discard the useless feature information, thus generating the feature maps $C_i \in \mathbb{R}^{H_i \times W_i \times c_i} (i = 1, 2, 3, 4, 5)$ for the self-attention process. The sizes of $C_i$ are $(H/2) \times (W/2)$, $(H/4) \times (W/4)$, $(H/8) \times (W/8)$, $(H/16) \times (W/16)$, and $(H/32) \times (W/32)$ ($H$ and $W$ are the height and width of the input images), respectively. The channel numbers of $C_i$ are 4, 16, 64, 256, and 1024, respectively.

*2) Self-Attention:* First, the feature maps $C_i \in \mathbb{R}^{H_i \times W_i \times c_i} (i = 1, 2, 3, 4, 5)$ need to be cut into patches of size $P$, $(P/2)$, $(P/4)$, $(P/8)$, and $(P/16)(P = 16)$, respectively. These patches can be mapped to different regions of the feature map from multiple scales. These patches are then subjected to flattening and location labeling operations to generate feature sequences for the self-attention process, as shown in the following equation:

$$S_i \in \mathbb{R}^{(P_i^2 \times c_i) \times \frac{H \times W}{P_i^2}}, \quad (i = 1, 2, 3, 4, 5) \tag{1}$$

where $S_i$ represents the feature sequences. $(H, W)$ is the dimension of the feature maps and $P_i$ is the size of the patches. $c_i$ represents the number of channels of $F_i$. $P_i^2 \times c_i$ represents the dimension of the feature sequences, and $(H \times W/(P_i^2))$ represents the number of feature sequences.

After the feature sequences are obtained, these feature sequences need to be linearly transformed to generate a vector representation for the self-attention process. The process of the linear transformation is shown in the following equation:

$$\begin{cases} Q_i(\text{query}) = S_i \cdot W_{\text{q}} \\ K_i(\text{key}) = S_i \cdot W_{\text{k}} \\ V_i(\text{value}) = S_i \cdot W_{\text{v}} \end{cases} \tag{2}$$

where $W_{\text{q}}$, $W_{\text{k}}$, and $W_{\text{v}}$ are all trainable matrices.

Then, use compression and splicing operations to splice $K_i \in \mathbb{R}^{N \times d} (i = 1, 2, 3, 4, 5)$ into a public key value $K_p \in \mathbb{R}^{N \times d}$, where $N$ represents the number of key values and $d$ represents its dimension. The specific values are $N = 196$ and $d = 1024$, and the $K_p$ generation process is shown in the following equation. The purpose of $K_i$ stitching is to fuse high-level features with low-level features, so that high-level features can guide the modeling process of low-level features [39], thus improving the ability of the transformer to capture boundary details

$$K_p = \phi\left[\gamma(K_i)\right] \otimes K_5 = \left(\frac{K_1 + K_2 + K_3 + K_4}{4}\right) \otimes K_5 \tag{3}$$

where $\phi$ represents the splicing operation, $\gamma$ represents the compression operation, and $\otimes$ represents the bitwise product operation.

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After completing the above steps, the self-attention process can be performed, which is shown in (4)

$$\mathrm{SA}_i(Q, K, V) = \alpha\left[\beta\left(\frac{Q_i \cdot K_p^{\mathrm{T}}}{\sqrt{d}}\right)\right] \cdot V_i \tag{4}$$

$$\text{MF-SA}_i = \frac{\left(\mathrm{SA}_i^1 + \mathrm{SA}_i^2 + \cdots + \mathrm{SA}_i^h\right)}{h} \tag{5}$$

$$\mathrm{Out} = \mathrm{MLP}[\alpha(\text{MF-SA}_i)] \tag{6}$$

where $\alpha(\cdot)$ represents the layer normalization operation, and $\beta(\cdot)$ represents the Softmax function. $h$ represents the number of heads, and MF-SA represents the output processed by multi-head self-attention. MLP represents the multilayer perceptron, which consists of a fully connected layer, a dropout layer, and a Gaussian Error Linear Units (GELU) [40] activation function.

### E. Decoder

The role of the decoder is to fuse the feature information from the MT and encoder. Then, the fused feature information is upsampled step by step to obtain a lake water segmentation image with the same size as the original image. In particular, to maintain the consistency of the feature dimension, the hidden feature sequence from the transformer needs to be reshaped to adjust the size from $(H \times W/(P_i^2)) \times c_i$ to $(H/P_i) \times (W/P_i) \times c_i$. In NT-Net, the decoder consists of five US_blocks. Each US_block contains a concatenate layer, a $1 \times 1$ convolution kernel, a Rectified Linear Unit (ReLU) activation function, and a $2\times$ upsampling operation.

### F. $\mathrm{Loss}_{seg}$ Loss Function

NT-Net uses the $\mathrm{Loss}_{\mathrm{seg}}$ loss function to train and optimize the node parameters of the neural network. The $\mathrm{Loss}_{\mathrm{seg}}$ loss function consists of a binary cross-entropy loss ($\mathrm{Loss}_{\mathrm{bce}}$) and a dice coefficient loss ($\mathrm{Loss}_{\mathrm{dice}}$), which can be defined as follows:

$$\mathrm{Loss}_{\mathrm{seg}} = \lambda_1 \mathrm{Loss}_{\mathrm{bce}} + \lambda_2 \mathrm{Loss}_{\mathrm{dice}} \tag{7}$$

where $\mathrm{Loss}_{\mathrm{bce}}$ represents the binary cross-entropy loss, and $\mathrm{Loss}_{\mathrm{dice}}$ represents the dice coefficient loss. $\lambda_1$ and $\lambda_2$ are the adjustable hyperparameters. In this article, we set $\lambda_1 = 1$ and $\lambda_2 = 1$.

The distribution of lake water samples in remote sensing images is extremely uneven, which makes the final training results dominated by negative samples with more pixels, resulting in inaccurate segmentation results. NT-Net uses Dice loss [41] for this problem, and its definition is shown in the following equation:

$$\mathrm{Loss}_{\mathrm{dice}}(P, G) = 1 - \frac{2\sum_{h=1}^{H}\sum_{w=1}^{W}|G_{h,w} \cap P_{h,w}|}{\sum_{h=1}^{H}\sum_{w=1}^{W}\left(|G_{h,w}| + |P_{h,w}|\right)} \tag{8}$$

where $P$ is the prediction result and $G$ is the ground truth in the label image. $W$ and $H$ represent the width and height of the image, respectively.

Dice is a region-dependent loss, and the loss of the current pixel is not only related to the predicted value of the current pixel, but also related to the predicted value of other points. When the proportion of positive samples in the image is small, Dice will overwhelm some negative samples, making training more inclined to mine foreground regions. However, when there are too many tiny targets, the convergence process is unstable, and even gradient saturation occurs in extreme cases. To optimize the training process of the network, we introduce a binary cross-entropy loss in NT-Net. Unlike Dice loss, binary cross-entropy loss is a pixelwise measure of the difference between the actual distribution and the predicted distribution. Since it directly utilizes the prediction of pixels to adjust the gradient, it is more stable to optimize the model's training process. Therefore, we use the cross-entropy loss to make up for the inadequacy of the Dice loss in model optimization, which is defined as shown in the following equation:

$$\mathrm{Loss}_{\mathrm{bce}}(P, G) = -\sum_{h=1}^{H}\sum_{w=1}^{W}\left[G_{h,w} \times \log P_{h,w} + \left(1 - G_{h,w}\right)\log\left(1 - P_{h,w}\right)\right]. \tag{9}$$

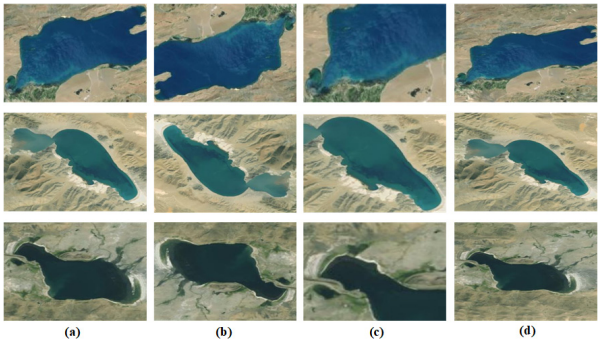



Fig. 6. Augmentation process of the dataset: (a) original images; (b) flipped; (c) randomly cropped; and (d) stretched.

## IV. Experiments

### A. Experimental Details

NT-Net is built in the Pytorch deep learning framework and trained using five NVIDIA RTX 3080 (8G) GPUs. To achieve fast convergence of the network, we use the Stochastic Gradient Descent (SGD) optimizer and set momentum 0.9 and weight decay 1e-4 to optimize the back-propagation process of NT-Net. We extracted 5283 remote sensing images of lakes with a size of $448 \times 448$ using Google Earth and randomly selected 4021 images as a training dataset and 1262 images as a testing dataset. To improve the generalization ability of the network model to changes in lake morphology, we use rotation, stretching, and random cropping to perform data augmentation on the training dataset as shown in Fig. 6. The augmented training dataset contains 8061 remote sensing images of lake water bodies.

We use the following evaluation metrics to compare the performance differences of different semantic segmentation networks: intersection over union (IoU), Dice (Dice similarity coefficient), and Acc, as shown in the following equations:

$$\mathrm{IoU} = \frac{1}{N}\sum_{n=1}^{N}\frac{\mathrm{TP}}{\mathrm{FP} + \mathrm{FN} + \mathrm{TP}} \times 100\% \tag{10}$$

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TABLE I

COMPARISON RESULTS OF NT-NET AND CLASSICAL SEMANTIC SEGMENTATION NETWORK

| Network | *IoU*/% | *Dice*/% | *Acc*/% |
|---|---|---|---|
| U-Net [42] | 65.63 | 71.89 | 74.51 |
| PspNet [43] | 70.10 | 73.65 | 76.02 |
| DeepLabV3+ [44] | 72.47 | 76.85 | 80.81 |
| **Our (NT-Net)** | **85.26** | **89.38** | **91.49** |

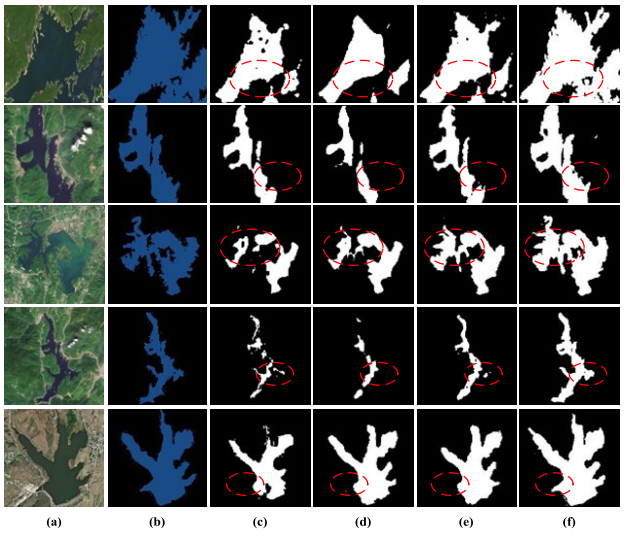



Fig. 7. Visual results of NT-Net and classical semantic segmentation networks: (a) images; (b) ground truth; (c) U-Net; (d) PspNet; (e) DeepLabV3+; and (f) NT-Net.

$$\text{Dice} = \frac{1}{N}\sum_{n=1}^{N}\frac{2\times \text{TP}}{\text{FP}+\text{FN}+2\times \text{TP}}\times 100\% \tag{11}$$

$$\text{Acc} = \frac{1}{N}\sum_{n=1}^{N}\frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{FN}+\text{FP}}\times 100\% \tag{12}$$

where $N$ is the number of remote sensing images, TP is the number of pixels classified as lakes, TN is the number of pixels classified as background, FP is the number of pixels misclassified as lakes, and FN is wrongly classified as the background number of pixels.

### *B. Comparison With Classic Semantic Segmentation Networks*

To ensure the fairness of the experiments, all networks use DS-block in the encoder. The experiments compare the performance differences between NT-Net and U-Net [42], PspNet [43], and DeeplabV3+ [44]. As shown in Table I, NT-Net achieves significant improvement on all evaluation metrics. The IoU of NT-Net is 85.26%, which is 12.79%–19.63% higher than other networks, which indicates that it is more accurate for lake localization and segmentation. The Dice and Acc of NT-Net are 89.38% and 91.49%, respectively, which are 12.53%–17.49% and 10.68%–16.98% higher than other networks, which means that NT-Net has better practical generalization ability in pixel recognition Acc.

Fig. 7 shows a visual comparison of NT-NET with other networks. As can be seen from Fig. 7, there are many information holes in the segmentation results of U-Net, because U-Net lacks an effective way to perceive the contextual information of images. Although the segmentation results of PspNet and DeepLabV3+ are relatively good, they lack the semantic judgment ability to deal with background interference, resulting in the inability to reproduce the boundary information of the lake. Compared with other networks, NT-Net is not only able to preserve the continuity of lakes, but also accurately segment tiny lakes at the boundaries. In the face of complex background information, NT-Net uses the IAM to suppress the feature performance of background information, making the boundary contour of the lake more refined. Based on the above analysis, it can be seen that NT-Net can accurately identify and segment lake water bodies in images and shows an all-around performance that is better than that of classical semantic segmentation networks.

TABLE II

COMPARISON RESULTS OF NT-NET AND MAINSTREAM LAKE SEGMENTATION NETWORKS

| Network | *IoU*/% | *Dice*/% | *Acc*/% |
|---|---|---|---|
| Lae-Net [45] | 75.86 | 77.64 | 81.57 |
| HA-Net [46] | 77.21 | 79.02 | 83.38 |
| SR-SegNet [47] | 82.05 | 83.25 | 85.46 |
| MFA-Net [14] | 80.95 | 82.49 | 85.12 |
| FEW-Net [21] | 82.83 | 84.33 | 87.25 |
| **Our (NT-Net)** | **85.26** | **89.38** | **91.49** |

### *C. Comparison With the Lake Segmentation Networks Based on Pure CNN Structure*

The experiments compare the performance differences between NT-Net and Lae-Net [45], HA-Net [46], SR-SegNet [47], MFA-Net [14], and FWE-Net [21]. As can be seen from Table II, NT-Net achieves the best values on all evaluation metrics. Compared with other networks, NT-Net improves IoU, Dice, and Acc by 2.43%–9.40%, 5.05%–11.74%, and 4.24%–9.92%, respectively. In the above data, the improvement of NT-Net on Dice is particularly obvious, which indicates that NT-Net can more accurately locate the specific location of lakes in the image.

Fig. 8 shows the visualization results of NT-Net and the comparison networks. As shown in Fig. 8, the networks based on pure CNN structure cannot accurately locate the boundary position of the lakes, resulting in low segmentation Acc of the lake boundary. For example, in the last row of Fig. 8, the contrastive networks ignore the boundary information of the middle region of the image. The reason for the above phenomenon is that the receptive field of these networks is small and cannot perceive the contextual information of the boundary. Compared with other networks, the lakes obtained by NT-Net not only have more coherent boundary contours, but also reproduce tiny lakes at the boundary. This is because the MT used by NT-Net can model the contextual distribution of boundary information from the global scope of the images, thus preserving the integrity of boundary contours. Since the MT can utilize high-level semantic features to assist the modeling process of detailed low-level features, NT-Net can further

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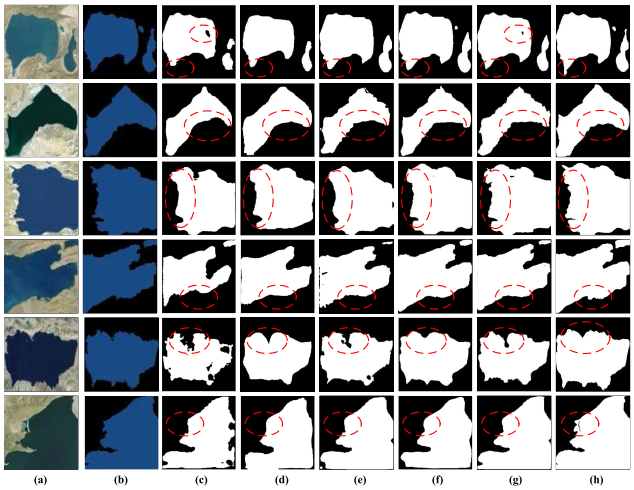



Fig. 8. Visual results of NT-Net and mainstream lake segmentation networks: (a) images; (b) ground truth; (c) Lae-Net; (d) HA-Net; (e) SR-SegNet; (f) MFA-Net; (g) FEW-Net; and (h) NT-Net.

TABLE III
COMPARISON RESULTS BETWEEN NT-NET AND SEGMENTATION NETWORKS BASED ON PURE TRANSFORMER STRUCTURE

| Network | *IoU*/% | *Dice*/% | *Acc*/% |
|---|---|---|---|
| BA-Net [30] | 78.01 | 82.44 | 84.33 |
| Swin-UNet [48] | 80.20 | 83.59 | 86.15 |
| TransUNet [37] | 78.89 | 82.57 | 85.35 |
| TransFuse [38] | 81.65 | 84.11 | 86.67 |
| STransFuse [31] | 82.13 | 85.97 | 88.52 |
| **Our (NT-Net)** | **85.26** | **89.38** | **91.49** |

refine the edge texture of lakes. Through the experiments in this section, it can be verified that NT-Net can produce more accurate results than the network models based on the pure CNN structure.

### *D. Comparison With the Segmentation Networks Based on Pure Transformer Structure*

The experiments compare the performance differences between NT-Net and BA-Net [30], Swin-UNet [48], TransUNet [37], TransFuse [38], and STransFuse [31]. It can be seen from Table III that NT-Net shows the best performance on the three evaluation indicators of IoU, Dice, and Acc. Compared with TransFuse, NT-Net improves IoU, Dice, and Acc by 3.61%, 5.27%, and 4.82%, respectively; compared with STransFuse, it improves by 3.13%, 3.41%, and 2.97%, respectively. At the same time, it is also significantly higher than BA-Net, SwinUNet, and TransUNet on all evaluation metrics. The above results show that NT-Net can produce better segmentation results than other network models in the task of segmenting lake water bodies.

Fig. 9 shows the visualized experimental results. From the fifth and sixth rows of Fig. 9, it can be seen that the contrastive networks incorrectly classify some background pixels into the lake category, which leads to the occurrence of the over-segmentation problem. This is because contrastive networks cannot suppress the feature representation of background information. Unlike other networks, NT-Net effectively alleviates the over-segmentation problem by using the IAM to weaken the adverse effects of background objects on the segmentation results. Apart from the over-segmentation problem, the contrastive networks cannot recover the detailed texture of the lake. For example, the segmentation results obtained by BA-Net are significantly different from the ground truth. The reason for the above problem is that BA-Net only performs self-attention processing on the feature maps of the high level, while ignoring the utilization of the low-level feature maps. Because the low-level feature maps contain a lot of feature information for scene interpretation, the network cannot accurately reconstruct the detailed features at the boundary. Compared with BA-Net, NT-Net can use self-attention modeling for multiple levels of features and utilizes high-level features to guide the modeling process of low-level features, resulting in a finer edge segmentation effect.

TABLE IV
RESULTS OF THE MT MODULE ABLATION EXPERIMENTS. "—" REPRESENTS THE ELIMINATION OF RELATED MODULES

| Network | *IoU*/% | *Dice*/% | *Acc*/% | *Recall*% |
|---|---|---|---|---|
| **NT-Net** | **85.26** | **89.38** | **91.49** | **93.82** |
| NT-Net—MT | 82.15 | 84.47 | 85.19 | 82.04 |

Through the experiments in this section, it can be proved that NT-Net can obtain more accurate segmentation results compared with the current transformer-based network model. In addition, by observing Figs. 7–9, it can be found that compared with other networks, NT-Net still obtains higher segmentation Acc when facing lakes with different shapes. This shows that NT-Net benefits from the advantages of model structure and can obtain stronger generalization ability under the condition of data augmentation.

### *E. Ablation Studies*

To evaluate the performance of the MT module and the IAM, ablation experiments on NT-Net are conducted. In the experiment, the Recall and Precision evaluation indicators are added to evaluate the effectiveness of each module more intuitively, as shown in following equations:

$$\text{Recall} = \frac{1}{N}\sum_{n=1}^{N}\frac{\text{TP}}{\text{FN}+\text{TP}} \times 100\% \tag{13}$$

$$\text{Precision} = \frac{1}{N}\sum_{n=1}^{N}\frac{\text{TP}}{\text{FP}+\text{TP}} \times 100\% \tag{14}$$

where $N$ is the number of remote sensing images, TP is the number of pixels classified as lakes, FP is the number of pixels misclassified as lakes, and FN is wrongly classified as the background number of pixels. The function of Recall is to evaluate the severity of the under-segmentation problem. The lower the value, the more serious the under-segmentation problem is. The role of Precision is to evaluate the severity of the over-segmentation problem. The lower the value, the more serious the over-segmentation problem is.

*1) Multilevel Transformer:* It can be seen from Table IV that without using the MT module, the network model has a relatively large reduction in the evaluation indicators of IoU, Dice, Acc, and Recall, decreasing by 3.12%, 4.91%, 6.30%,

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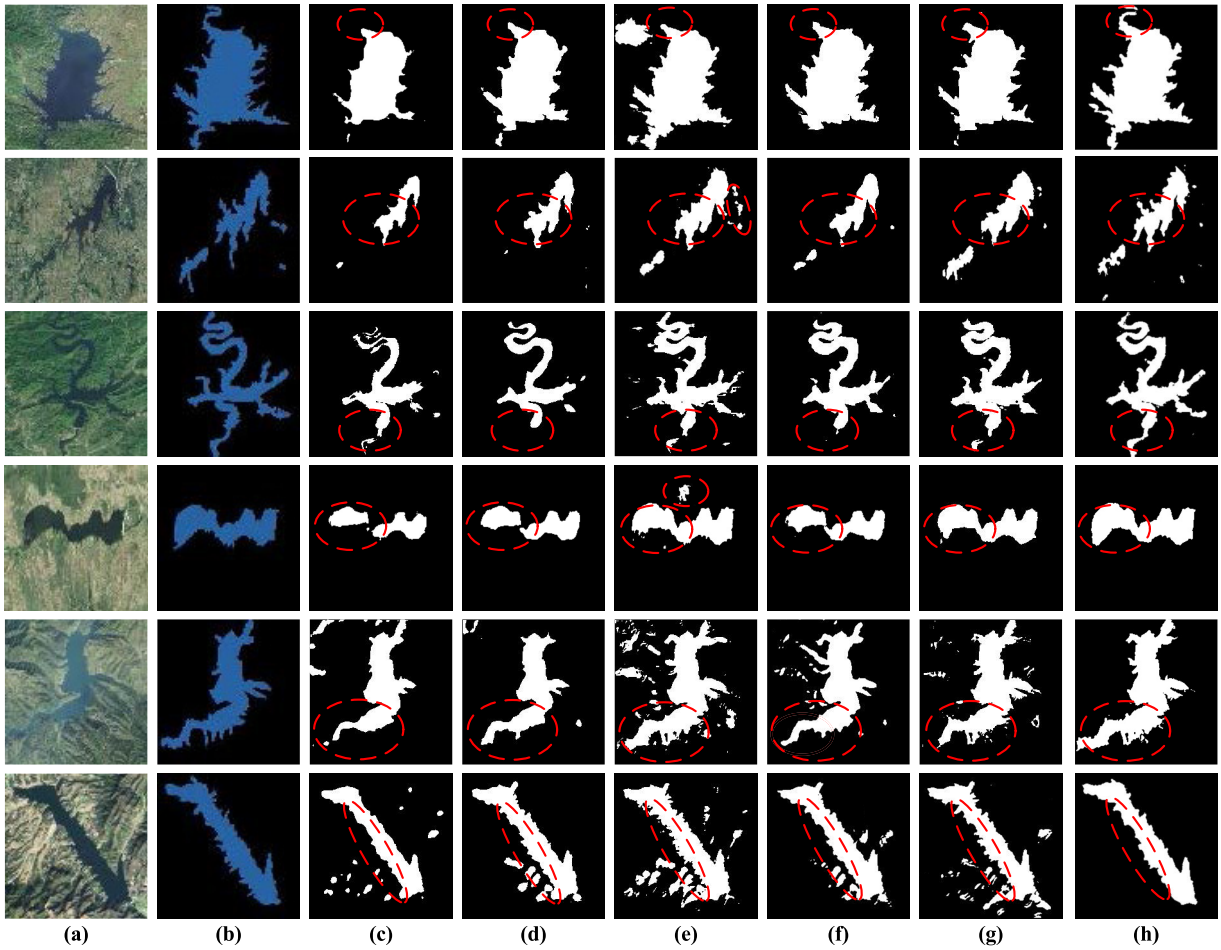



Fig. 9. Visualized segmentation results for the comparative experiments: (a) images; (b) ground truth; (c) BA-Net; (d) Swin-UNet; (e) TransUNet; (f) TransFuse; (g) STransFuse; and (h) NT-Net.

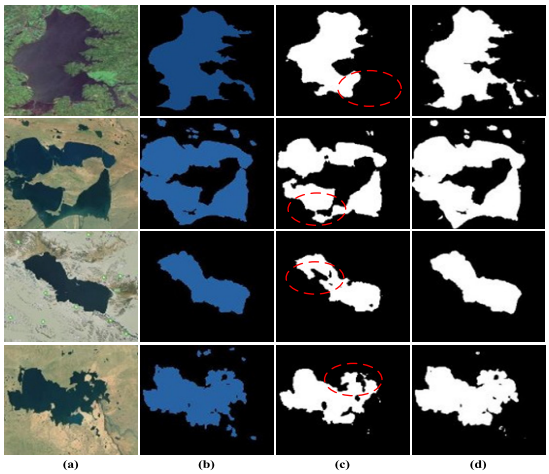



Fig. 10. Visualization results of the MT ablation experiment: (a) images; (b) ground truth; (c) NT-Net-MT; and (d) NT-Net.

and 11.78%, respectively. Among them, the decrease in the Recall value is the most obvious. The above results show that the lack of MT module will cause NT-Net to fail to accurately locate lake water bodies in images, resulting in under-segmentation problems.

Fig. 10 shows the visualization results of the experiment. As shown in Fig. 10, when the MT module is missing, many information holes appear at the boundary of the lake, destroying the integrity of the boundary contour. This is because NT-Net can only use ordinary convolution to aggregate the contextual information of local regions after the lack of the MT module. Since fragmented local context information cannot restore the distribution of boundary information in global space, it destroys the integrity and continuity of boundary contours. After using the MT module, the boundary of the lake water body is clearer and the outline is more complete. This is because MT can capture the semantic dependencies of objects from a global perspective and refine the modeling results of detailed low-level features. Through the experiments in this section, it can be verified that the MT module can show performance advantages to be trusted in improving the integrity of the lake water boundary.

TABLE V
RESULTS OF THE IAM ABLATION EXPERIMENTS

| Network | *IoU*/% | *Dice*/% | *Acc*/% | *Precision*/% |
|---|---|---|---|---|
| Baseline | 80.52 | 83.37 | 84.43 | 84.62 |
| Baseline + DAM | 82.05 | 85.91 | 87.74 | 88.51 |
| Baseline + RAM | 82.96 | 86.63 | 87.25 | 89.12 |
| **Baseline + IAM** | **85.26** | **89.38** | **91.49** | **93.29** |

*2) Interference Attenuation Module:* In this section, we conduct ablation experiments on the IAM, as well as the DAM

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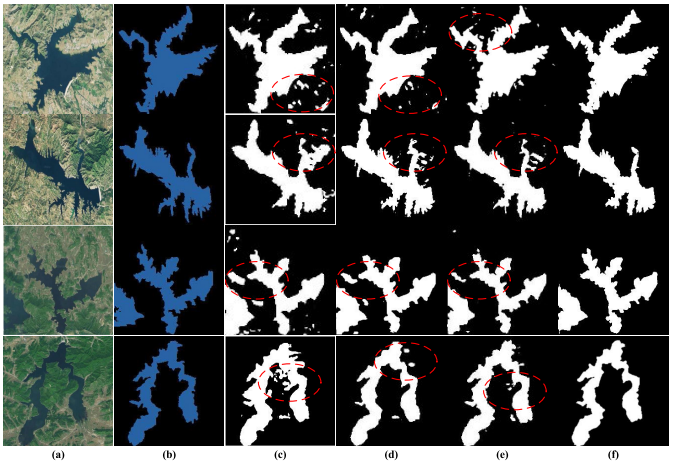



Fig. 11. Visualized segmentation results of the IAM ablation experiment: (a) images; (b) ground truth; (c) baseline; (d) baseline + DAM; (e) baseline + RAM; and (f) baseline + IAM.

and RAM contain. As can be seen from Table V, the baseline network with the IAM removed has the worst performance. Compared with the baseline network, the network using DAM or RAM showed significant improvement in IoU, Dice, Acc and Sen evaluation metrics. This shows that the DAM and RAM used in IAM can effectively suppress the feature representation of nonlake objects. Compared with other networks, the network using IAM achieves the best performance in terms of IoU, Dice, Acc and Sen evaluation indicators, and the improvement of Precision is the most obvious. This result shows that the IAM can improve the anti-interference ability of the network, thus solving the problem of over-segmentation in the segmentation results.

Fig. 11 shows the visualization results of the experiment. As can be seen from Fig. 11(c), when the IAM is removed, some background regions in the result are incorrectly segmented, especially at the boundary of the lake water body, there is an obvious over-segmentation problem. When the network uses DAM or RAM, although the problem of over-segmentation is effectively alleviated, the effect is not obvious. This is because interfering objects not only have lake-like characteristics, but also have some unique positional association with surrounding pixels. Therefore, the feature difference between lakes and disturbing objects cannot be modeled using only positional changes or channel associations. When the network uses the IAM, the over-segmentation problem is effectively alleviated. This is because IAM can take the advantage of the differences in deep semantics between lakes and other ground objects, thus effectively suppressing the feature representation of interfering objects and giving the target more discriminative feature weights. The experiments in this section demonstrate that the module's ability to suppress background interference is reliable.

*3) Discussion of the $\lambda_1$ and $\lambda_2$ in Loss$_{seg}$:* $\text{Loss}_{\text{seg}}$ uses two hyperparameters $\lambda_1 = 1$ and $\lambda_2 = 1$ to balance the advantages and disadvantages of Dice loss and binary cross-entropy loss. To verify the rationality of the hyperparameter settings in this article, the impact of different $\lambda_1$ and $\lambda_2$ on the network segmentation Acc is compared. The training and testing process of the experiment uses remote sensing images with a positive sample ratio of less than 35% to compare the experimental results more intuitively, as shown in Fig. 12.

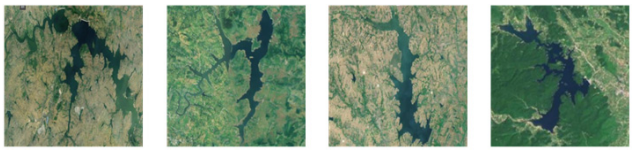


Fig. 12. Remote sensing images of lake water bodies with a positive sample ratio of less than 35%.

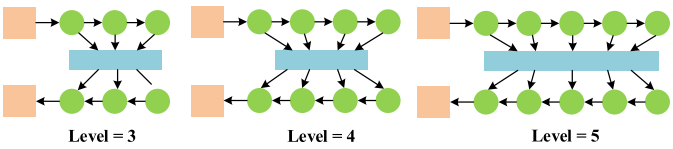



Fig. 13. Schematic of NT-Net at different sampling layers.

It can be seen from Table IV that when $\lambda_2$ representing Dice loss is 0.5 and 0.7, the segmentation Acc of the network is low, especially when $\lambda_2$ is 0.5, the IoU and Acc of the network have dropped significantly by 3.22% and 5.67%, respectively. The above phenomenon shows that when the number of foregrounds and background pixels is seriously unbalanced, the decrease of the Dice ratio will hinder the training process of the network and reduce the fitting degree of the parameters to the samples. When $\lambda_1$ representing the cross-entropy loss is 0.5 and 0.7, the segmentation Acc of the network is also lower. This is because the cross-entropy loss can maintain a high gradient state, improve the stability of the network convergence process, and update the parameters in a direction conducive to accurate segmentation. Therefore, the proportional decrease of the cross-entropy loss will affect the final segmentation Acc of the network. This article also compares the cases of $\lambda_1 = 1$, $\lambda_2 = 0.9$ and $\lambda_1 = 0.9$, $\lambda_2 = 1$, and finds that the segmentation Acc of the network differs very little. Therefore, considering comprehensively, we set $\lambda_1$ and $\lambda_2$ to be 1 uniformly.

## V. Discussion

### A. Network Depth of the NT-Net

NT-Net performs five downsampling operations and five upsampling operations on the input image, respectively. Its purpose is to enable the network to fully extract the semantic features of lake water without generating too many redundant features. To verify the rationality of the above settings, we compared NT-Net with 3, 4, 5, and 6 sampling layers (as shown in Fig. 13) and verified the variation of the Acc (Dice) with the number of iterations. To facilitate the training of NT-Net with six layers, the dataset size used in the experiment is $512^2$.

As shown in Fig. 14, when the sampling layers of NT-Net are 3 and 4, the verification Acc is low because the shallow network cannot fully extract the feature information in the image, which limits the performance of the network. When

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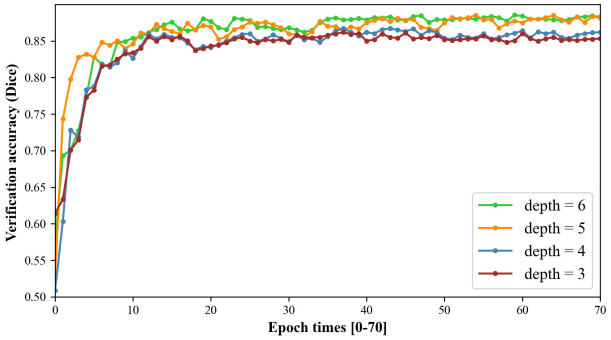



Fig. 14. Effect of the number of sampling layers on the validation Acc.

TABLE VI
EFFECT OF DIFFERENT $\lambda_1$ AND $\lambda_2$

| *Parameters* | *IoU*/% | *Acc*/% |
|---|---|---|
| $\lambda_2 = 1,\ \lambda_1 = 0.7$ | 81.24 | 85.36 |
| $\lambda_2 = 1,\ \lambda_1 = 0.5$ | 80.97 | 83.94 |
| $\lambda_1 = 1,\ \lambda_2 = 0.7$ | 80.46 | 82.73 |
| $\lambda_1 = 1,\ \lambda_2 = 0.5$ | 79.35 | 81.05 |
| **$\lambda_1 = 1,\ \lambda_2 = 1.0$** | **82.57** | **86.72** |

TABLE VII
EFFECT OF NETWORK LAYERS

| **Depth** | ***IoU*/%** | ***Dice*/%** | ***Acc*/%** |
|---|---|---|---|
| Layer=3 | 82.69 | 85.03 | 88.75 |
| Layer=4 | 84.24 | 85.58 | 89.63 |
| **Layer=5** | **85.26** | **89.38** | **91.49** |
| Layer=6 | 85.48 | 89.07 | 92.12 |

the number of sampling layers of NT-Net is 6, its verification Acc is almost the same as that of NT-Net with five layers, and both are higher than NT-Net with depths 3 and 4. However, a too deep network will lead to an increase in computational complexity and also produce redundant features. To further demonstrate the impact of the difference in the number of sampling layers on the segmentation Acc, we compared NT-Nets with sampling layers of 3, 4, 5, and 6, and the results are shown in Table VII.

As shown in Table VII, when the number of sampling layers is 5, the network can show certain advantages in various evaluation indicators. This proves that when the number of sampling layers is 5, NT-Net can give full play to its performance and produce more accurate segmentation results.

### *B. Number of Layers of the MT*

This article sets up an eight-layer MT module in NT-Net. This setting expands the attention relationship between pixels, thus enhancing the global modeling ability of NT-Net for images. To verify the rationality of the above settings, we conducted the following experiments: using the training set data to train NT-Nets with 4, 6, 8, 10, and 12 layers of MT modules, respectively. Then, the change of IoU value with the number of iterations during network model training was compared (every 20 epochs are counted). The experimental results are shown in Fig. 15.

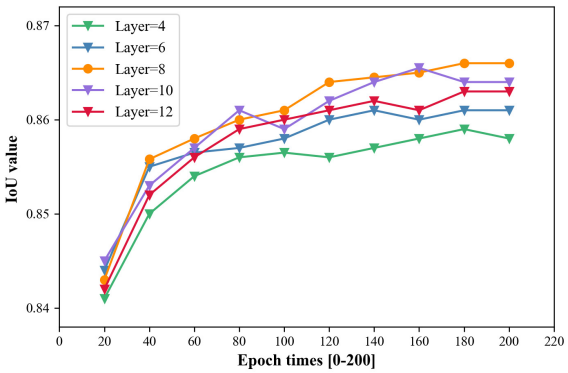



Fig. 15. IoU values vary with the number of transformer layers.

TABLE VIII
EFFECT OF INPUT RESOLUTION

| Input size | ***IoU*/%** | ***Dice*/%** | ***Acc*/%** | ***Time*/min** |
|---|---|---|---|---|
| 384×384 | 83.72 | 86.51 | 88.30 | 693 |
| **448×448** | **85.26** | **89.38** | **91.49** | **1157** |
| 512×512 | 86.31 | 89.75 | 91.60 | 1734 |

It can be found from Fig. 15 that when the number of MT modules is 8, NT-Net can achieve the highest segmentation Acc. The explanation for this phenomenon is that the eight-layer MT module helps to model accurate node features. In addition, setting up an eight-layer MT also helps NT-Net to expand the spatial connection of feature elements in the context area. In conclusion, if the number of layers is too small, it will reduce the ability of the network to capture nonlocal semantic dependencies. If there are too many layers, it will cause much redundant computation, hindering the optimization process of the network model. Therefore, from the perspective of algorithm operation efficiency, we set up an eight-layer MT module in NT-Net to produce more advantageous segmentation results.

### *C. Effect of Input Resolution*

In NT-Net, we set the input resolution to 448 × 448. To verify the rationality of this setting, we test results with 384 × 384, 448 × 448, and 512 × 512 resolutions as NT-Net input is given, as shown in Table VIII.

As shown in Table VI, while keeping the number of patches fixed, the segmentation Acc of NT-Net increases with the input resolution. Taking the Dice value as an example, the segmentation Acc of NT-Net gradually increased from 86.51% to 89.75%. Although the segmentation Acc of NT-Net improves slightly when the input resolution size is increased from 448 × 448 to 512 × 512, the training time of the network increases significantly. Therefore, considering the computational overhead and the running efficiency of the algorithm, this article sets the input resolution of NT-Net to 448 × 448.

### *D. Model Complexity of the NT-Net*

To demonstrate that the excellent performance of NT-Net is due to its efficient structural advantage rather than the huge number of parameters, the following experiments are set up.

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TABLE IX

COMPARISON OF DIFFERENT NETWORKS IN TERMS OF PARAMETER SIZE, TRAINING TIME, AND ACC. THE ENCODER PARTS OF BOTH U-NET AND DEEPLABV3+ USE THE DS-BLOCK DESIGNED IN THIS ARTICLE. (CNN: U-NET, DEEPLABV3+ TRANSFORMER: BA-NET, STRANSFUSE)

| **Network** | Input size | *IoU*/% | Parameters/M | Time/min |
|---|---|---|---|---|
| U-Net | 448×448 | 65.63 | 36.54 | 544 |
| DeepLabV3+ | 448×448 | 72.47 | 68.42 | 735 |
| BA-Net | 448×448 | 78.01 | 115.91 | 1084 |
| STransFuse | 448×448 | 81.65 | 108.32 | 973 |
| **Our** | **448×448** | **85.26** | **127.06** | **1127** |

The experiments compared U-Net, DeepLabV3+, BA-Net, STransFuse, and NT-Net in terms of parameter size, model size, training time, and IoU. The results are shown in Table IX.

As shown in Table IX NT-Net is significantly larger than CNN-based network models in terms of model parameters and model size. The reason is that the execution of the transformer involves complex matrix operations. Compared with transformer-based BA-Net, the number of parameters of NT-Net has increased by 11.15M, and the segmentation Acc has increased by 7.25%. Compared with STransFuse, the number of parameters of NT-Net has increased by 18.74M, and the segmentation Acc has increased by 3.61%. The above results show that although NT-Net achieves higher Acc, it does not lead to a large increase in model parameters and training time. Unlike BA-Net, which has a large number of parameters and low Acc, NT-Net achieves a good tradeoff between model complexity and segmentation Acc despite its complex model structure. Therefore, the experiments in this section can prove that the excellent performance of NT-Net mainly depends on the effectiveness of modules.

## VI. CONCLUSION

In this article, we design a semantic segmentation network NT-Net that fuses transformer and CNN for the accurate extraction of lake water bodies in remote sensing images. NT-Net takes full advantage of the inductive bias ability of CNN in modeling information correlation and the modeling ability of transformer in global context information. Benefiting from the effectiveness of the structure, NT-Net can not only extract rich semantic features from parallel multipaths containing multiple levels, but also effectively express the contextual connections of semantic features. We use remote sensing images of lakes extracted from Google Earth to test the performance of NT-Net and other semantic segmentation networks. The experimental results show that NT-Net can obtain segmentation Acc better than other classical semantic segmentation networks, which proves the advancement and effectiveness of NT-Net. In addition, we also verify the excellent performance of the IAM and the MT module through ablation experiments. Overall, our study provides a new method for accurate and efficient extraction of lake water bodies from remote sensing images. Because of the high labor cost for the production of high-quality labels, we will try to transform NT-Net into a semisupervised model-based semantic segmentation network in future work. The network can extract lake water bodies in remote sensing images by using a small number of training samples.

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